# Day 14 | Tuesday

November 24, 2026

14

## Measurement systems and readable dimensions

12:45 PM–2:00 PM • Math • 75 min

## Exact conversion factors and mixed-unit practice

2:15 PM–3:30 PM • Math • 75 min

**Week 5 • 2 scheduled blocks • 150 teaching minutes**

Estimate. Show the method. Label units. Check the result.

# Today’s exact schedule

## 12:45 PM–2:00 PM • 75 min

### Measurement systems and readable dimensions

| | | |
|---|---|---|
| 12:45 PM–12:50 PM | **Start** | 5 |
| 12:50 PM–12:55 PM | **Recall** | 5 |
| 12:55 PM–1:10 PM | **Teach** | 15 |
| 1:10 PM–1:25 PM | **Together** | 15 |
| 1:25 PM–1:45 PM | **Practise** | 20 |
| 1:45 PM–1:55 PM | **Check** | 10 |
| 1:55 PM–2:00 PM | **Wrap up** | 5 |

Math • Math Week 5, Lesson 1

## 2:15 PM–3:30 PM • 75 min

### Exact conversion factors and mixed-unit practice

| | | |
|---|---|---|
| 2:15 PM–2:20 PM | **Start** | 5 |
| 2:20 PM–2:25 PM | **Recall** | 5 |
| 2:25 PM–2:40 PM | **Teach** | 15 |
| 2:40 PM–2:55 PM | **Together** | 15 |
| 2:55 PM–3:15 PM | **Practise** | 20 |
| 3:15 PM–3:25 PM | **Check** | 10 |
| 3:25 PM–3:30 PM | **Wrap up** | 5 |

Math • Math Week 5, Lesson 2

**Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time**

# Measurement systems and readable dimensions

**Today’s goal: Read metric and imperial lengths and convert within one system before using mixed-unit drawings.**

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

Exercise Book 7A Q1–10 (p.45); 5A Q3 (p.32)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Recall and reconnect

Use Exercise Book 7A Q1,7; ask which quantities are lengths and which are capacities.

**Attempt the retrieval task independently; explain the method before checking.**

Resources: Exercise Book 7A Q1–10 (p.45); 5A Q3 (p.32)

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Recall from the Exercise Book | 7A Q1

Convert 4.75 m to mm.

Write the operation, working and units before checking.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Convert within the metric system

**1** 1 m = 100 cm = 1,000 mm; 1 cm = 10 mm.

**2** Moving to a smaller unit makes the numerical count larger.

**3** Moving to a larger unit makes the numerical count smaller.

**4** Write the starting and ending units beside the conversion.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Read feet, inches and drawing labels

1 ft = 12 in; convert the feet before adding the extra inches.

To reverse the conversion, group complete sets of 12 inches.

Read the tape’s graduation before naming a fractional inch.

Use written drawing dimensions; do not measure the displayed picture.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Change the unit—not the length

1 m = 100 cm = 1,000 mm
2.35 m = 235 cm = 2,350 mm

**1 ft = 12 in**
**5 ft 7 in = 5 × 12 + 7 = 67 in**

Together: 7A Q2–3 and Q8–10. Read units aloud.
Check 67 in by dividing into groups of 12: 5 whole feet and 7 in.

Numeracy Hub: Measuring Tools and Unit Conversions.

# Textbook | page 110

| Unit | Symbol | Equals |
|---|---|---|
| inch | in or ″ | 1/36 yard |
| foot | ft or ′ | 12 in = 1/3 yard |
| yard | yd | 3 ft = 36 in |
| mile | mi | 1,760 yd = 5,280 ft |

**Table 7.2** Imperial length units. Source: Weights and Measures Act (Canada), Schedule II, Measurement of Length.

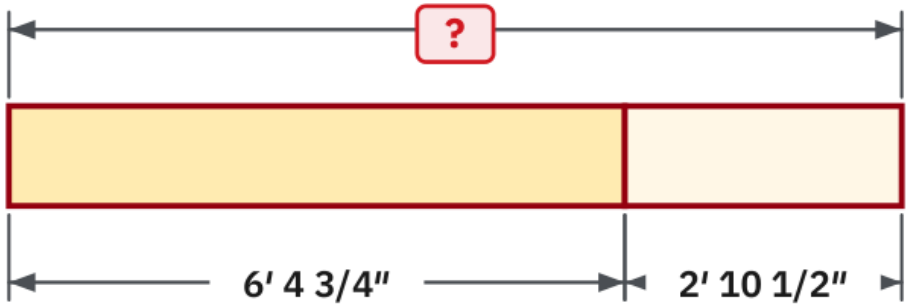
Textbook p.110 • Table 7.2 — Imperial length units

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Textbook | page 110

## Example 7.3 Adding feet and inches

Two pieces of 6′ 4 3/4″ and 2′ 10 1/2″ are laid end to end. Find the total length.




**SOLUTION**

1. Add the feet: 6 + 2 = 8 ft. Add the inches: 4 3/4 + 10 1/2 = 15 1/4 in.
2. 15 1/4 in is more than a foot: 15 1/4 − 12 = 3 1/4 in, and carry 1 ft.
3. Total: 8 + 1 = 9 ft and 3 1/4 in.
4. Check in inches: 76.75 + 34.5 = 111.25 in = 9′ 3 1/4″.

Textbook p.110 • Example 7.3 — Adding feet and inches

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Spot the mistake. Explain the correction.

## A length is written in feet and inches. Can the two numbers simply be added?

Discuss the method before checking the answer.

**Convert the feet to inches first, then add the remaining inches in the same unit.**

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Work together

Work Exercise Book 7A Q2–3,8–10. Work with a partner to read and say the units aloud.

**Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.**

Resources: Exercise Book 7A Q1–10 (p.45); 5A Q3 (p.32)

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Guided practice | 7A Q2

Convert 625 mm to m.

Work with a partner. Take turns explaining the next step.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Compare your method | 7A Q2

Convert 625 mm to m.

**CHECK TOGETHER**

**0.625 m**

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Your scheduled practice

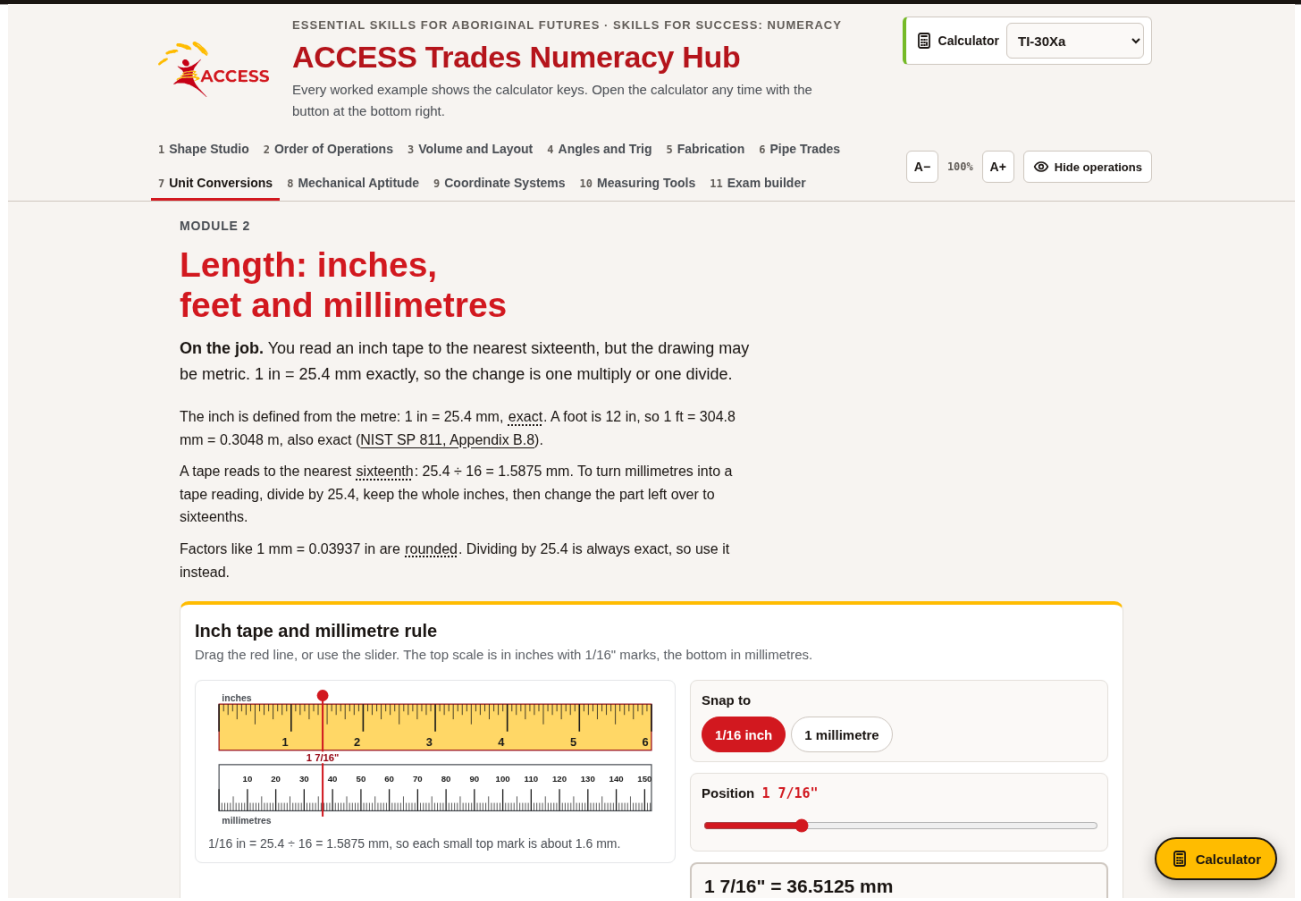
Complete remaining 7A Q1–10 and 5A Q3. Measure two classroom objects with rule/tape; record units and compare readings with a partner.

**Solve on paper first; check with tools; correct one error and explain the change.**

Resources: Exercise Book 7A Q1–10 (p.45); 5A Q3 (p.32)

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Open Numeracy Hub




Navigation example. Use the values and units in the assigned task.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Practise in Numeracy Hub and Shape Canvas

1. Use Numeracy Hub Measuring Tools to read a rule and a tape before checking.

2. Use Unit Conversions to check the assigned within-system conversion.

3. Read a Shape Canvas rectangle in mm and in from its labels, not its screen size.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Practise from the Exercise Book | 7A Q1

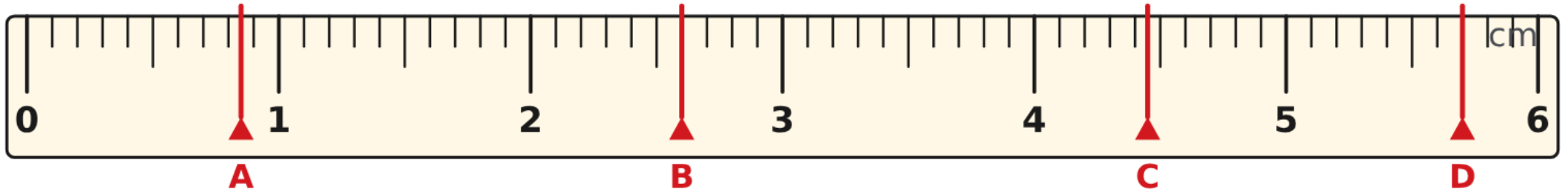
Convert 4.75 m to mm.

Write the operation, working and units before checking.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Practise from the Exercise Book | 5A Q3

Read marks A, B, C and D to the nearest 0.5 mm.




Write the operation, working and units before checking.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Check independently

Exercise Book 7A Q8–9; show conversion of mixed feet/inches and a reverse check.

**Solve without displayed answers; include working, units and final rounding.**

Resources: Exercise Book 7A Q1–10 (p.45); 5A Q3 (p.32)

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Independent check | 7A Q8

Write 9′ 7 3/4″ in inches.

Answer independently. Keep worked solutions closed.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Independent check | 7A Q9

Write 94 in in feet and inches.

Answer independently. Keep worked solutions closed.

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

**Name the math and one other essential skill used; record a next practice step.**

Resources: Exercise Book 7A Q1–10 (p.45); 5A Q3 (p.32)

Textbook sections 7.1–7.2 pp.108–110; sections 4.2 pp.57–58

# Break

2:00 PM–2:15 PM

Return ready for: Exact conversion factors and mixed-unit practice

**This break is excluded from teaching minutes.**

# Exact conversion factors and mixed-unit practice

**Today’s goal: Convert inch and metric dimensions using the book factor, rounding only to the requested final precision.**

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

Exercise Book 7A Q11–16 (pp.45–46); 7B Q1–2 (p.47); 7C Q1–3 (p.49)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Recall and reconnect

Discuss whether converting inches to millimetres should enlarge or shrink the number. Retrieve 1 in = 25.4 mm.

**Attempt the retrieval task independently; explain the method before checking.**

Resources: Exercise Book 7A Q11–16 (pp.45–46); 7B Q1–2 (p.47); 7C Q1–3 (p.49)

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Use a conversion factor with direction

1. 1 in = 25.4 mm is an exact length conversion.
2. Inches to millimetres: multiply by 25.4.
3. Millimetres to inches: divide by 25.4.
4. Write the factor so the original unit cancels and the target unit remains.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Separate exact values from rounded readings

1. Keep the exact factor and full calculation precision.

2. Predict whether the numerical result should increase or decrease.

3. Round only the final result to the requested unit or tape graduation.

4. Convert back before rounding to check the conversion direction.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Arrange the factor to cancel the unit

3/4 in × 25.4 mm/in = 19.05 mm
50 mm ÷ 25.4 mm/in ≈ 1.968504 in

**Nearest 1/16 in: 0.968504 × 16 ≈ 15.496 → 15 sixteenths.**
**Rounded measurement = 1 15/16 in.**
**The exact factor is not the same as the rounded result.**

Together: 7A Q11–14. Convert one answer back before rounding.
Numeracy Hub Unit Conversions; Shape Canvas for 7C Q1.

# Textbook | page 111

### INCHES AND MILLIMETRES

$$mm = in \times 25.4$$

$$in = mm \div 25.4$$

1 in = 25.4 mm exactly (NIST SP 1038, section 5.1.3). It also follows from the Weights and Measures Act: 0.9144 m ÷ 36 = 0.0254 m.

Textbook p.111 • Exact inch–millimetre conversion factors

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Textbook | page 111

## Example 7.4 A fraction of an inch in millimetres

A drawing calls for 3/8 in plate. What is that in millimetres?

**SOLUTION**

1. Inches to millimetres: multiply by 25.4.
2. 3/8 × 25.4 = 0.375 × 25.4 = 9.525 mm.

Textbook p.111 • Example 7.4 — Fractional inches to millimetres

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Spot the mistake. Explain the correction.

## A learner divides an inch measurement by 25.4 to get millimetres. Does the size make sense?

Discuss the method before checking the answer.

**No. One inch contains 25.4 millimetres, so the numerical count must increase; multiply.**

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Work together

Solve Exercise Book 7A Q11–14 together; check one by converting back before rounding.

**Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.**

Resources: Exercise Book 7A Q11–16 (pp.45–46); 7B Q1–2 (p.47); 7C Q1–3 (p.49)

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Guided practice | 7A Q11

Convert 5/8 in to mm.

Work with a partner. Take turns explaining the next step.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Compare your method | 7A Q11

Convert 5/8 in to mm.

**CHECK TOGETHER**

**15.875 mm**

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Your scheduled practice

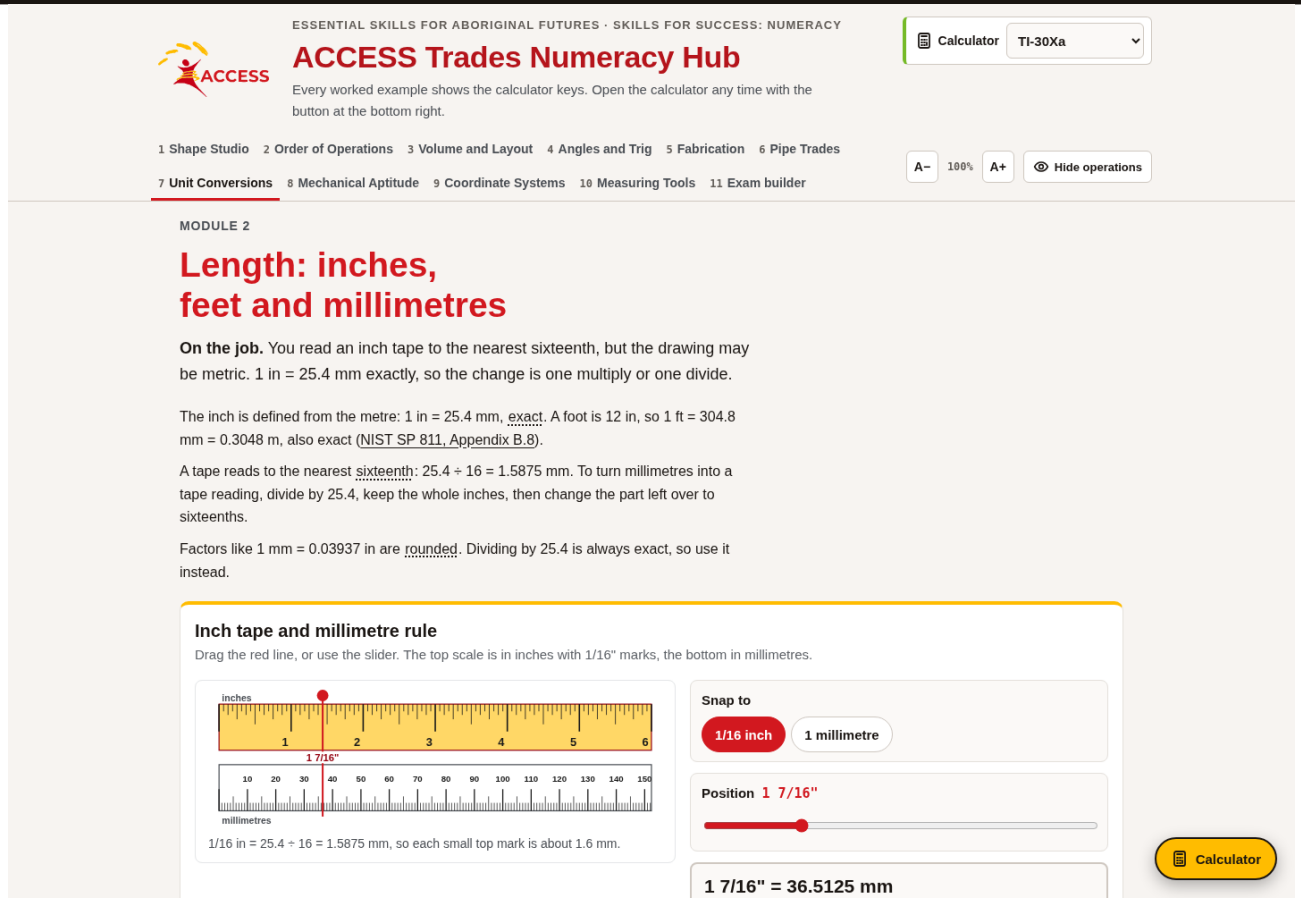
Complete Exercise Book 7B Q1–2 and 7C Q1–3. Show the factor and units on paper; use the Numeracy Hub and Shape Canvas only after predicting the result.

**Solve on paper first; check with tools; correct one error and explain the change.**

Resources: Exercise Book 7A Q11–16 (pp.45–46); 7B Q1–2 (p.47); 7C Q1–3 (p.49)

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Open Numeracy Hub




Navigation example. Use the values and units in the assigned task.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Practise in Numeracy Hub and Shape Canvas

1. Predict and calculate each assigned conversion on paper with the factor and units.

2. Check a length conversion in Numeracy Hub Unit Conversions.

3. Build the 7C Q1 rectangle in Shape Canvas; switch displayed units and compare labels.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Practise from the Exercise Book | 7B Q1

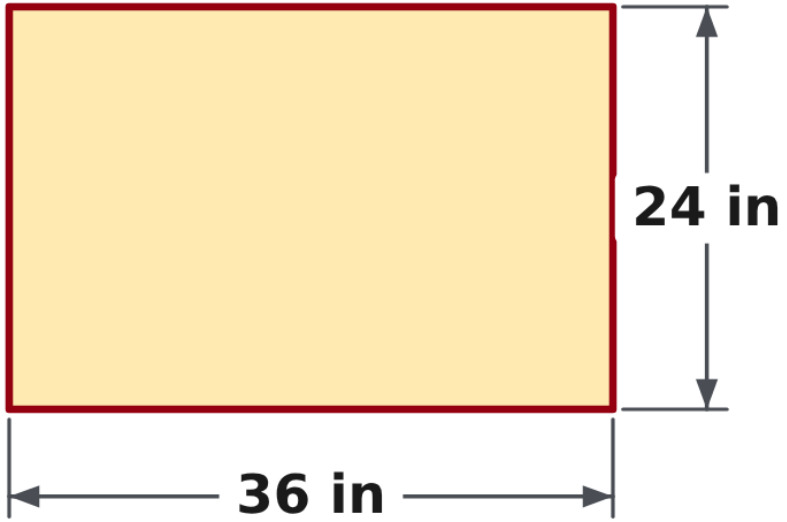
A metric drawing gives a length of 2,235 mm. Your tape reads in feet and inches. What do you mark, to the nearest 1/16 in?

Write the operation, working and units before checking.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Practise from the Exercise Book | 7C Q1

Write both sizes of the sheet in millimetres.




Write the operation, working and units before checking.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Check independently

Exercise Book 7A Q11 and Exercise Book 7B Q1; show the exact result before final tape rounding.

**Solve without displayed answers; include working, units and final rounding.**

Resources: Exercise Book 7A Q11–16 (pp.45–46); 7B Q1–2 (p.47); 7C Q1–3 (p.49)

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Independent check | 7A Q11

Convert 5/8 in to mm.

Answer independently. Keep worked solutions closed.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Independent check | 7B Q1

A metric drawing gives a length of 2,235 mm. Your tape reads in feet and inches. What do you mark, to the nearest 1/16 in?

Answer independently. Keep worked solutions closed.

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221

# Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

**Name the math and one other essential skill used; record a next practice step.**

Resources: Exercise Book 7A Q11–16 (pp.45–46); 7B Q1–2 (p.47); 7C Q1–3 (p.49)

Textbook sections 7.3 pp.111–112; conversion factors pp.220–221